

VM-22 Life Contingent Annuity Rate Calculator (Non-Jumbo Contracts)

Purpose

This tool calculates quarterly statutory maximum valuation interest rates under NAIC VM-22 for life-contingent, non-jumbo annuity contracts, based on Treasury yields, Table X (spreads), Table A (defaults), and prescribed weightings.

Key Formula

$$I_q = R + S - D - E$$

Where:

- R = Reference Rate (weighted average of U.S. Treasury yields)
- S = Spread Rate (weighted average from Table X by WAL and credit rating)
- D = Default Cost (weighted average from Table A by WAL and credit rating)
- E = Spread Deduction (fixed at 0.25%)

Final quarterly valuation interest rate:

$$\text{VM22 Rate} = \text{round}[(R + S - D - E) * 4] / 4$$

Inputs Required

1. Year (e.g., 2024)
2. Quarter (e.g., 2)

Followed by file uploads:

- Treasury Yields: CSV with "Date", "2 Yr", "5 Yr", etc.
- Table X: Spread rates by WAL and Moody's rating
- Table A: Default cost rates by WAL and Moody's rating
- Reference Rate Weights
- Spread Rate Weights
- Default Cost Weights
- Daily Corporate Rates (optional)

Class: `VM22\_Calculator`

Initialization

```
```python
calc = VM22_Calculator(year_input=2024, quarter_input=2)
```
```

Method: `reference\_rate()`

Calculates the reference rate using weighted Treasury yields.

Method: `spread\_rate()`

Calculates the weighted average spread from Table X.

Method: `default\_cost()`

Calculates the weighted default cost from Table A.

Method: `VM22\_Rates()`

Combines R, S, D and subtracts 0.0025, then annualizes and rounds.

Expected Output

| Quarter | Bucket | VM22 Quarterly Rate |
|---------|-----------|---------------------|
| 2024Q2 | 0-4 Years | 1.25% |
| 2024Q2 | 5-9 Years | 1.50% |
| 2024Q2 | 10+ Years | 1.75% |

Regulatory Notes

- Non-Jumbo Contracts: <\$250M per plan, per insurer, within 90 days.
- Reference Period: Quarter prior to valuation quarter.
- Deduction: Fixed at 0.25% (0.0025 quarterly).
- Rounding: Nearest multiple of 0.25%.

Assumptions and Limitations

- Only for life contingent, non-jumbo products.
- Assumes proper CSV formatting.
- No UI beyond file uploads.
- Future work: Jumbo handling, discount factor calc, visualization.